

# Institutions Matter: Economic freedom and income mobility

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## ABSTRACT

Intergenerational income mobility — the ability of children to rise above their parents' economic status — is a key concern in debates over inequality and opportunity. Most research has focused on structural factors like social capital, overlooking the role of economic freedom (market-oriented institutions, lower regulation, secure property rights). Using data on U.S. metropolitan areas, we show that higher economic freedom is strongly associated with greater income mobility. Children in high-freedom areas experience 5 %–12 % more upward mobility than those in low-freedom areas. Economic freedom and social capital operate independently, suggesting multiple paths to mobility. These results highlight that reforms aimed at expanding economic freedom can directly improve opportunity — a practical lever for policymakers concerned with improving income mobility.

Since the early 2000s, a growing portion of economic debates has focused on the broad study of inequality. Closely intertwined with these discussions is social mobility, particularly income mobility, due to the connection between high inequality and reduced upward social mobility — a link confirmed across multiple studies (Corak, 2013; Durlauf et al., 2022).

One of the mechanisms explaining this link is the social capital of individuals near the top of the income distribution. These individuals often benefit from dense social networks that provide privileged access to information, opportunities, and influence, reinforcing their socio-economic position (and that of their children). This form of social capital, known as “bonding” social capital, contributes to the persistence of relative status over time (Browne, 2005; Mitnik et al., 2024). In contrast, other forms of social capital—referred to as “bridging” social capital, or

more recently as “economic connectedness” in the network literature—can mitigate this persistence. By granting access to influential networks, valuable information flows, and career-enhancing opportunities, bridging social capital facilitates upward mobility for individuals with limited financial resources (Chetty et al. 2014, 2022a, 2022b; Levy et al., 2020; Gurzo et al., 2024; Jackson, 2025; Fang et al., 2025).

This common emphasis in the literature has tended to overshadow the role of more immediate economic policies and institutions in offsetting the persistence of income status.<sup>1</sup> More precisely, the potential role of economic freedom—defined by secure property rights, limited regulation, open trade, sound monetary policy, and limited government spending—in fostering income mobility has received comparatively less attention.<sup>2</sup> In this paper, we build on the limited literature examining the relationship between economic freedom and intergenerational

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<sup>1</sup> With the exception of policies related to the provision of education (see more in the next footnote).

<sup>2</sup> For example, the works of Chetty et al. (e.g., 2018a; 2018b; 2022a; 2022b) generally invoke “markets” (such as “labor markets conditions”) as exogenous factors that are commonly shared with little discussions of how institutions shape those “conditions. When the word “institution” appears in those works, it is generally to refer to “educational institutions” (Chetty et al. 2020). The best we could find was in Chetty et al. (2014) where appendix H (pp. 58–59) deals briefly with an associational study of local public goods and targeted tax credits with mobility. No consideration is given to broader aspects of taxation and government spending or regulatory environments.

income mobility by asking whether economic freedom facilitates (or depresses) upward mobility across generations, and how its effects compare to those of social capital, which has been extensively studied.

Using data from Chetty et al. (2022a, b) on social capital and intergenerational income mobility in the United States, combined with subnational economic freedom data from 1972 to 2017 (Stansel, 2019), we employ the same associational econometric strategy to ensure comparability. Our analysis reveals several key findings: a) economic freedom—both in aggregate form and its individual components—significantly enhances absolute mobility (income gains relative to parents) and relative mobility (income rank relative to parental rank); b) economic freedom shows no association with racial mobility gaps; c) economic freedom's effects match those of “bridging” social capital (which promotes income mobility) while overpowering “bonding” social capital (which depresses income mobility). Using interaction terms, we find that d) economic freedom and “bridging” social capital operate independently, suggesting additive effects; e) “bonding” social capital and economic freedom function as substitutes; and f) economic freedom's effects on mobility are direct rather than mediated through reduced inequality. To address potential endogeneity concerns, we employ synthetic instruments (Lewbel instruments) and find that our results remain robust.<sup>3</sup> To address problems tied with false rejections of the null hypothesis resulting from multiple hypothesis testing (when using economic freedom and its components), we employ the Holm-Bonferroni procedure. Our pattern of statistical significance is largely unaffected.

Fig. 1 below summarizes as succinctly as possible elements a) through e) of our results. The figure shows the effect of going up one quartile in economic freedom with and without controlling for economic connectedness (i.e., the variable most privileged by Chetty et al., 2022a). The bottom quartile in each is set as reference category. As can be seen, going up from the bottom to the top quartile in economic freedom without controlling for economic connectedness is tied to 11.6 % and 4.6 % higher levels of absolute and relative mobility (there are no effects on the racial gap). In contrast, when controlling for economic connectedness, these figures are respectively 9.5 % and 4.5 %.

In addition to assessing the relative importance of economic freedom and social capital with respect to social mobility, our results also produce a noticeable enhancement to the existing literature on economic freedom and income mobility. It looks at intergenerational income mobility (rather than intragenerational which misses parts of the social capital aspects) at the subnational level in the United States, which has relatively high-quality institutions – notably property rights protection – by international standards. The earlier literature having found that a sizable share of the cross-country differences in intergenerational income mobility stem from property rights protection, our results show that even within a country with well-established property rights, variations in economic freedom continue to significantly affect intergenerational mobility outcomes. Economic freedom benefits seem to extend beyond more “basic” aspects of institutions to include regulatory burden, government spending and taxation rates.

Our paper is organized as follows: Section 2 explains why economic freedom could matter to income mobility. Section 3 explains our empirical strategy and data. Section 4 presents the results. Section 5 discusses and concludes by highlighting how future researchers need to develop causal inference approaches to test the importance of economic freedom on improving income mobility.

<sup>3</sup> In a supplementary file, we provide multiple alternative specifications (e.g., swapping variables, LASSO regressions, limiting years of economic freedom data, using alternative measures of pro-market institutions, using measures of social connectedness found within other published studies, etc.) that confirm these findings.

## 1. Social capital, economic freedom and income mobility

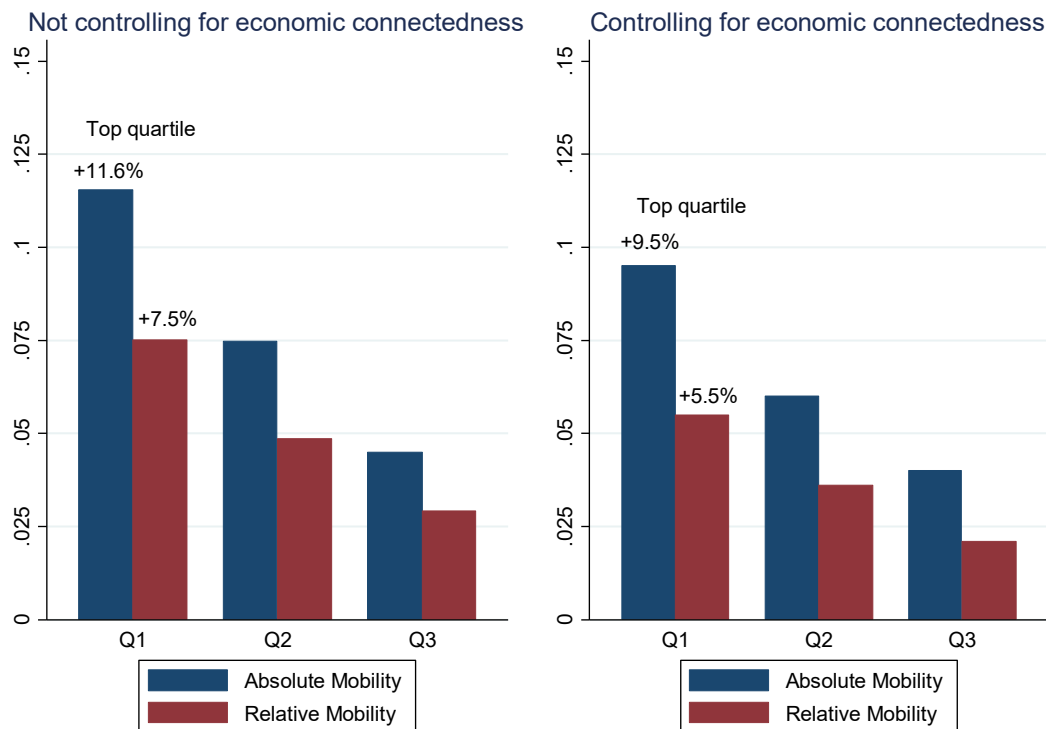
Social capital has a connection to income mobility – both across generations or within one's lifetime (Güell et al., 2018; Chetty et al., 2022a). This can be tied to “structural” views of social mobility whereby birth-related or early-life conditions create systemic advantages or disadvantages. Each rung of the socio-economic ladder (or set of rungs) can be thought of as having a social network. The value of network membership being a function of the wealth of its members, children born into affluent families are in networks that unlock more opportunities for them as adults (Bourdieu, 2013; Zimmerman, 2019). Social networks composed of members at the bottom of the ladder offer less attractive opportunities reflecting the constraints imposed by less wealth (Chantarat and Barrett, 2012). Overall, this leads to structural reproduction of inequality which is ultimately visible in limited income mobility (especially “relative” mobility where we compare people's centile position in the distribution relative to their parents). It is this structural reproduction mechanism that gives rise to the often-discussed “Great Gatsby Curve” showing high inequality means low intergenerational income mobility (Corak, 2013; Durlauf et al., 2022).

Homophily – the tendency to associate with social kin – explains this but not in full. Indeed, social networks can span across multiple rungs of the ladder in ways that weaken the structural reproduction mechanism (implying imperfect homophily). Chetty et al. (2022a) are particularly enlightening in discussing how this can be the case. There are different types of social capital – bridging and bonding – the former is most important. Equating “bridging” with “cross-type connectedness”, Chetty et al. (2022a, p. 109) argue that the “extent to which different types of people (for example, high income versus low income) are friends with each other” is the most important determinant of income mobility in the United States. “Bonding” social capital (which they label as “network cohesiveness” because of the degree to which networks of contacts are clustered into cliques) as well as measures of community-level characteristics of civic engagement that do not speak to the depth and types of connection between people are far weaker determinants of intergenerational income mobility. They can even – as per the structural reproduction argument – depress mobility. Phrased differently, “bonding” social capital can serve to increase the costs of outsider attempts to enter the network.<sup>4</sup> “Bridging” social capital, on the other hand, creates social networks that can span multiple socio-economic groups (thereby reducing the interaction cost between people of different income ranks).<sup>5</sup>

The extensive focus on social capital in mobility research has led to many omitting the role of economic freedom. To understand this role, we must first define “economic freedom.” According to Gwartney and Lawson (2003, pp. 406–7), the institutions that promote economic freedom protect property rights, leave exchanges largely voluntary, place no barriers to competition and promote personal choice. “To provide an infrastructure for voluntary exchange, and protect individuals and their property from aggressors seeking to use violence, coercion, and fraud to seize things that do not belong to them” (p. 406), governments must secure property rights, avoid excessive taxation

<sup>4</sup> The frequently-invoked French sociologist Pierre Bourdieu would speak of codes (or habits) that are costly to master and learn but allow to screen non-members pretending to be members to get the benefit of the network. These “codes” impose prohibitive signaling costs on outsiders who rationally remain in their own network.

<sup>5</sup> Staying within Bourdieu's framework, “bridging” social capital dilutes the exclusivity of network-related codes and habits which weaken the ability of high-income networks to screen out lower-income individuals. This can happen in a variety of ways: greater ease of observing the code by non-members, undermining of code enforcement by members who may value interaction with non-members, weaker perceptions of legitimacy of such codes leading to the network's atrophy, etc.



**Fig. 1.** Effect of Economic Freedom (with and without controls for economic connectedness) expressed as the advantage in mobility over the first quartile of economic freedom of each.

Source: See Tables 2 and 4

inclusive of taxation made indirectly via inflation (i.e., having sound money). Government must also “refrain from actions that interfere with personal choice, voluntary exchange, and the freedom to enter and compete in labor and product markets” (p. 407) via international trade barriers, regulatory hurdles, and excessive taxation that discourage effort. This is why international economic freedom indexes are generally constructed using five components: size of government, sound money (i.e., inflation), property rights protections, regulations and the degree of trade openness. When subnational indexes are used, there are only three components: government spending, taxation and regulation. This reflects the more limited variation in subnational variations within countries that could be tied to trade and monetary policy (we explain these indexes in more details in the next section, for now it suffices that we consider them abstractly).

This definition and the resulting measurements can also be connected to key aspects of the structural reproduction argument. A subset of the literature that uses economic freedom indices finds that higher scores with respect to economic freedom has significant associations with measures of social capital (Berggren and Jordahl, 2006; Bjørnskov, 2007; Bidner and Francois, 2011; Berggren and Nilsson, 2013; McCannon et al., 2018; Storr and Choi, 2019; Callais et al., 2022). The direction of the association depends on the type of social capital. For example, some argue that economic freedom and social capital are substitutes because social capital of the “bonding” type (rather than bridging) favors rent-seeking which reduces economic freedom (via, for example, weaker property rights, greater regulation, more tariffs etc.) and in turn economic growth (Berggren et al., 2008; Williamson and Mathers, 2011; Mathers and Williamson, 2011; Holcombe, 2018; Graafland, 2020; Forrester and Nowrasteh, 2023). As a result, social networks could be strategically constructed to facilitate rent-seeking. Thus, they only reflect the underlying institutional structure that both creates exclusionary networks and limits economic opportunity. In this case, the structural reproduction problem changes since opening markets and eliminating legal barriers to entry (i.e., increasing economic freedom) diminishes social reproduction by making exclusive networks less

economically relevant. By increasing economic freedom (implying a reduction in rent-seeking), the mix of social capital should change from “bonding” towards “bridging” (Carilli et al., 2008; Meadowcroft and Pennington, 2008; Balamoune-Lutz, 2011).<sup>6</sup> Ultimately, by promoting this change in the mix of social capital, increases in economic freedom would promote intergenerational income mobility.

There is handful of works that draw a direct link between economic freedom and income mobility (or its proxies), by pointing to the role of economic freedom that enable market expansion and, in turn, upward mobility (Boudreaux, 2014; Dean and Geloso, 2022, 2024, Callais and Geloso, 2023; Callais et al., 2024).<sup>7</sup> This work identifies

<sup>6</sup> It may even offset “bad” social capital. Kapur et al. (2010) point out that lower castes in India saw the disappearance of many discriminatory caste practices post-liberalization. The reforms destroyed some localized social capital which was geared towards enforcing caste conformity. Thus, the liberalization essentially allowed for greater mobility of those at the bottom thanks to the destruction of “counterproductive” social capital (see notably Narayan and Pritchett, 1999; Jackson, 2017).

<sup>7</sup> Some studies make indirect connections. Meehan et al. (2019) and Geloso et al. (2024) found that occupational licensing of low-skill jobs affects intergenerational mobility in the U.S. Feldmann (2017) links economic freedom to human capital accumulation. Callais et al. (2024) show that economic freedom boosts intergenerational educational mobility. Alesina et al. (2021) find a connection in Africa between mobility and proximity to capitals, which may relate to property rights. However, a Google Scholar search for “income mobility” and “property rights” yields no other direct empirical studies, though secure property rights are widely seen as essential for mobility. Most literature focuses on structural factors, redistribution, and policies aimed at reducing immobility (Huang et al., 2021; Ichino et al., 2011; Pekkarinen et al., 2017; Chen et al., 2015; Rauh, 2017; Liu and Song, 2022). Even when positive links from economic growth to mobility are found (e.g., Neidhöfer, 2019), connections to property rights, market openness, and regulations are overlooked, despite a strong causal inference literature linking growth to these institutions (Billmeier and Nannicini, 2013; Grier and Grier, 2021; Callais and Young, 2023).

mobility-enhancing mechanisms that operate independently of social capital, complementing the social network dynamics discussed above. These mechanisms should be understood within the framework of the seminal work of [Becker and Tomes \(1979\)](#) (restated and expanded in [Becker et al. \[2018\]](#)). In this model, income mobility is a function of parental investment in their children's human capital. In the presence of imperfect capital markets and/or borrowing restrictions, poorer families opt for suboptimal investments in human capital. In contrast, richer families can fully finance the optimal investment level. In other words, the mere existence of high returns to human capital does not guarantee income mobility, as children of poorer families may remain disadvantaged, even in a meritocratic system.

However, “imperfection” in capital markets can result from weak property rights regimes—a key component of economic freedom ([Alchian and Demsetz, 1973](#); [Olson, 1993](#); [Avom et al., 2024](#)). The same logic applies to high inflation: if poorer households have more limited access to inflation-shielded assets, inflation reduces their ability to fund human capital investments, further entrenching the intergenerational transmission of socio-economic status. Trade tariffs may have a similar effect if they disproportionately increase the cost of goods consumed by lower-income families (thereby reducing their real incomes more than those of average households), while simultaneously benefiting higher-income families involved in protected industries.

These three components underscore a key limitation in the literature on economic freedom and income mobility. In developed countries, secure property rights, independent central banks, and broadly open trade regimes are largely settled features of the institutional environment. Consequently, there is relatively little cross-country variation in these components of economic freedom among high-income nations. Yet, [Boudreaux \(2016\)](#) and [Callais and Geloso \(2023\)](#) show that these components explain a substantial share of global variation in income mobility. This suggests that their explanatory power diminishes within richer countries, precisely because such countries have already converged on these institutional features, leaving limited scope for identifying marginal effects. This point is critical because, even within high-income countries, there exist substantial subnational differences in income mobility that remain to be explained ([Chetty et al., 2014](#); [Rothwell and Massey, 2015](#); [Corak, 2020](#); [Acciari et al., 2022](#); [Kenedi and Sirugue, 2023](#); [Connolly and Haeck, 2024](#)).

The only component that [Callais and Geloso \(2023\)](#) found that mattered heavily was regulatory burden and property rights protections. By erecting barriers to entry and reducing real incomes, depressed relative income mobility.<sup>8</sup> The complementary role of regulations they found was roughly as strong as the role of the security of property rights. The works of [Dean and Geloso \(2022, 2024\)](#) partly circumvent the limitations of the international comparisons. Both look at intragenerational income mobility for Canadian provinces since the 1980s. They find that taxation and labor market regulations had strong and consistent depressing effects on income mobility (especially from the lower end of the distribution).<sup>9</sup> The problem is that they were using *intra*-generational estimates which potentially misses the structural reproduction of inequality. Thus, while they can move to the heterogeneity of income mobility within nations, it comes at the conceptual cost of missing some of the structural reproduction logic. Shifting to

subnational *intergenerational* estimates of income mobility – which is possible for the United States – resolves this trade-off and thus we can build on these earlier works exploring economic freedom's role.

Before proceeding, we should point out that the components of economic freedom may not all move in the same direction. There are two sources of potential ambiguity discussed in the earlier literature, and one additional source—relevant to the present paper—that has not yet been mentioned.

First, the previously unmentioned source of ambiguity. There is a rather vibrant research field relating economic freedom to income inequality. That literature shows that economic freedom boosts income in all income groups but may do so unevenly ([Callais and Young, 2023](#); [Lawson and Dean, 2021](#)). A similar finding is made regarding economic freedom's effect on income across racial ([Hoover et al., 2015, 2018](#)) and gender ([Hall et al., 2020](#)) lines. As such, some articles find that inequality rises with economic freedom while other finds the reverse ([Ashby and Sobel, 2008](#); [Apergis et al., 2014](#); [Bennett and Nikolaev, 2017](#); [Bennett and Vedder, 2013](#); [Sturm and De Haan, 2015](#); [Bennett, 2024](#)). Given the negative relationship between income inequality and income mobility, mediation or amplification is possible. If economic freedom increases (reduces) income inequality, it could subsequently reduce (increase) income mobility. This possibility has never been formally considered – even within the emerging literature connecting economic freedom to mobility (e.g., [Callais and Geloso, 2023](#)).

Second, not all components of the economic freedom indexes are expected to go in the same direction. Union density, used as a sub-component for the labor market regulation component of subnational indices<sup>10</sup>, could adversely affect mobility if it is tied to lower employment levels and less investment in a given area. By raising wages above the marginal product of labor, high union density leads to lower employment and discourages investment, resulting in fewer jobs in the future. In the long run, compulsory union bargaining harms the economy ([Reed, 2003](#)). However, some could point that unions also perform a redistributive function such that they level inequality in ways that could promote intergenerational income mobility ([Farber et al., 2021](#); [Callais et al., 2025](#)). While the only available study linking union-related legislation (right to work laws that prohibit closed shop arrangements) shows that lower union density is tied to greater mobility ([Lilley, 2023](#)), the effect of unionization-related regulations is more ambiguous.

Third, the size of government, the last component (which is usually split into two parts for subnational indexes – spending and taxation) is most ambiguous. Higher levels of taxation, by reducing the returns to human capital investments, could depress upward mobility ([Feldmann, 2017](#)). They could also discourage effort and work, which in turn would reduce earnings and later income mobility. However, redistribution (captured in government spending) can reallocate resources to lower-income households, which can use them to make educational and business investment decisions that propel them upwards ([Lindert, 2021](#)). As such, we expect that the first two components might yield inconsistent results. This is also what [Dean and Geloso \(2022, 2024\)](#) found: government spending had no significant effect, while taxation had a small and significant adverse effect on mobility. Spending and taxation components effectively fought against each other. This left the labor market regulation component with the lion's share of the overall beneficial results on income mobility. However, if the disincentive effect of taxation on effort, labor, and savings is stronger (or weaker) than the pushing-up effects of transfers, then there is a net negative (or positive) effect on income mobility.

## 2. Data and empirical methods

The goal being to assess the relative strength of economic freedom

<sup>8</sup> Regulations can reduce the ability of poor households to finance human capital investments if the they reduce their real incomes in an uneven manner (either directly on income or indirectly via prices) ([Chambers et al., 2019](#)).

<sup>9</sup> Most notably, high minimum wages (a subcomponent of labor market regulation) were tied to reduced employment in lower income groups ([Dean and Geloso, 2024](#)). A result consistent with works on minimum wages and exits from poverty ([Sabia and Burkhauser, 2010](#); [Sabia and Nielsen, 2015](#)). In supplementary materials, we also replicated all our results with *all* the sub-components (this is 370 specifications that we ran). Minimum wages had strong adverse on income mobility measures.

<sup>10</sup> The assumption being that union density proxies for labor market laws forcing collective bargaining and unionization.



and the different types of social capital discussed in Chetty et al. (2022a), we must anchor our empirical strategy to theirs in order to preserve comparability. The main starting point of our analysis are the three mobility measurements of Chetty et al. (2022a) who use adjusted gross income data estimates of income mobility across generations. The first measure is one of “absolute mobility” which considers the “average income percentile rank in adulthood of children [born between 1978 and 1983] who grew up in that county with parents at the 25th percentile of the national parental household income distribution” (Chetty et al. 2022, p. 113). A person born in San Francisco but who now resides elsewhere is still assigned to San Francisco in that definition (this will matter as we will see below when we define how to include the MEFL). Because the starting point is so low in the distribution, any gains in rank relative to one’s parents imply the presence of absolute gains. The second measure is one of “relative mobility” which is the “slope from OLS regression of child rank on parent rank” (Chetty et al., 2014, datafile). This measure, being estimated on the entire population, assesses how much of a person’s position on the income ladder is determined by his parent’s income. On that measure, a smaller number is sign of less persistence/greater mobility. For interpretability, we invert the relative mobility scores in our results so that higher scores correspond to greater mobility. The third measure is the difference in absolute income mobility between Whites and Blacks. This is meant to assess whether economic freedom might have a differentiated effect across groups.<sup>11</sup> All the measures of income are based on tax returns and the adjusted gross income (AGI) which is a close cousin to market income (Chetty et al., 2022a).<sup>12</sup>

The measures of social capital also come from Chetty et al. (2022a) for the purposes of being as conceptually similar as they are and permit comparison. Their indicators are broken into three categories: cross-type connectedness, network cohesiveness and civic engagement. Cross-type connectedness is tied to “bridging” capital in the sense that it serves as “connector between otherwise disparate groups” (Jackson, 2020, p. 315). Network cohesiveness speaks to “bonding” in that it “describes connections *within* [emphasis is ours] a group or community characterized by high levels of similarity in demographic characteristics, attitudes, and available information and resources” (Institute for Social Capital, nd). The last category, which are “based purely on individual or community-level characteristics” (Chetty et al., 2022a, p. 109), can speak to bond “bonding” and “bridging”.

Chetty et al. (2022a) use data from social media to identify networks of friends. For cross-type connectedness, the variable is “economic connectedness” which is defined as the “share of high or above median income friends among people with low or below-median incomes” (Social Capital Atlas, nd). For network cohesiveness category, there are two variables: “clustering” and “support ratio”. The former is the rate at which two friends of a given person are in turn friends with each other. The latter captures the rate at which pairs of friends in a community have other friends in common. Finally, they use measures of local civic organizations and volunteering to speaks to the civic engagement. All these variables – organized at commuting zone-level or county-level – will be used for our baseline analysis.

Of all the variables they use (and even after using a LASSO procedure), Chetty et al. (2022a, pp. 114–15) argue that only economic connectedness is strongly and positively associated with economic

mobility. The others are weaker correlates – especially with respect to civic engagement. This latter result may be puzzling for some and it could be argued that this results from the civic engagement measures being less than ideal. To deal with this possibility, we (in an online appendix) replicate our results for civic engagement using the US Senate’s Social Capital Project (Social Capital Project, 2018). It contains a county-level index of social capital using 10 separate indicators that come sources different than those of Chetty et al. (2022a) and include measures of family unit and institutional trust. The pattern of correlation between these variables of civic engagement and the cross-type connectedness and network cohesiveness variables of Chetty et al. (2022a) is the same as within the latter’s work. However, for conciseness, we will only refer to replications in-text and refer readers to the supplementary materials.<sup>13</sup>

The economic freedom data (labeled MEFL) are taken from Stansel (2019) which are available at the five-year mark from 1972 to 2017 for 385 US metropolitan areas (henceforth MSA). Fig. 2 shows the map of the areas contained in the MEFL. This index is constructed on the same architecture as the commonly used index of Economic Freedom of North America (known as EFNA) produced by the Fraser Institute (Stansel et al., 2022). International indexes of economic freedom include property rights, monetary policy and trade policy. However, for subnational indexes, these components are absent.<sup>14</sup> Instead, there are three key components in subnational indexes: government spending (often called Area 1)<sup>15</sup>; taxation (Area 2)<sup>16</sup> and; labor market freedom (Area 3).<sup>17</sup>

The empirical literature on economic freedom commonly uses the components of economic freedom separately to shed light on the specific channels through which it influences outcomes. This is in large part because components have different effects. As indicated above when discussing the existing literature on economic freedom and income mobility, such uneven effects *do* exist and must be considered (Dean and Geloso, 2022, 2024; Callais and Geloso, 2023). There is also the possibility that they are multiplicative such that, individually, each component might have a small effect but – combined – have a larger effect (Bolen and Sobel, 2020; Dutta and Sobel, 2021). This possibility justifies examining both the individual components and the aggregate index to capture the full range of their potential interactions and combined influence.

We arrayed the Chetty et al. (2022a) data – which is at the commuting zone (CZ) level – at the MSA-level so that they would be consistent with the MEFL dataset. Within the primary data, provided by Chetty et al. (2022a), commuting zone statistics are assigned to the largest county in the territory. We aggregate these counties to the MSA-level by using population weights and cross-walking the county observations to the MSA. We only use MSAs in which there is a substantial overlap with the commuting zone, which eliminates many of the smallest MSAs from the sample. We are left with gap information for 261 medium and large MSA which are then matched to data from both the

<sup>13</sup> The number of robustness checks we conducted produced nearly 450 separate regressions in addition to those reported in this paper.

<sup>14</sup> Murphy (2020) built a rule of law measurement (i.e., property rights security) for a single year at the state-level. In our supplementary materials, we replicate our results but swap the state dummy variables for the state-level property rights measure constructed by Murphy. In the appendix, we also use occupational licensing data from Carpenter et al. (2021) to examine whether the results obtained using the economic freedom index—particularly its labor market regulation component—are conceptually mirrored by this alternative measure.

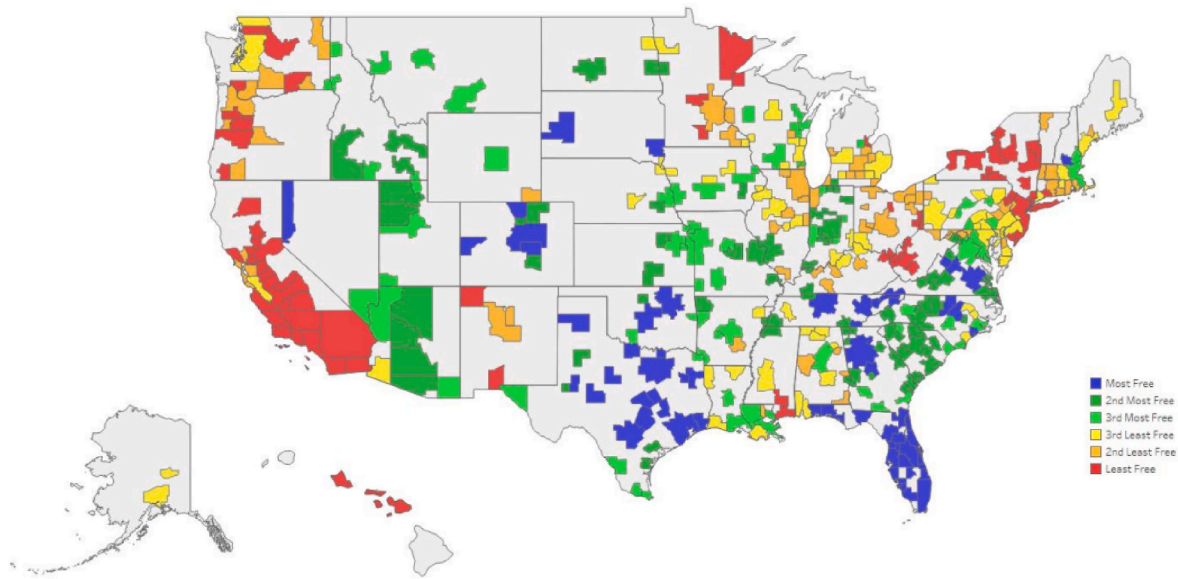
<sup>15</sup> Constructed on measures of government consumption, transfers and subsidies, social insurance and retirement payments.

<sup>16</sup> Based on data on income and payroll taxes, sales tax revenue, revenue from property tax, and tax revenue from each source except severance taxation (since this is levied at the state level only).

<sup>17</sup> Based on the local minimum wages, government employment shares and private union density.

<sup>11</sup> Which is consistent with the possibility of differential effects by racial groups raised by Hoover et al. (2015, 2018).

<sup>12</sup> AGI includes wages, tax-exempt interest income, non-taxable portions of social security and disability benefits if the parent files a tax return. This precludes potential issues of mechanically-driven effects (see Berggren, 1999) with some components (e.g., taxes) of our index of economic freedom. Moreover, the spending component of our index is measured on local government spending not federal (i.e., social security and disability benefits are thus not affecting the index).



**Fig. 2.** Map of the MEFI, 2017 for all the MSA available in the dataset.  
Source: Stansel (2019).

MEFI. The MEFI contains data from 1972 to 2017 at the MSA-level. While many precautions were taken to assure similar boundaries throughout the sample, the counties and boundaries of some MSAs have changed over time, with the most recent substantial change in the mid-1990s. There is not current crosswalk or weighted adjustment that can identify and remove this shift, but since we are using averages over many decades, we believe this does not have an impact on the reliability of our results.

Because Chetty et al. (2022a) use people born between 1978 and 1983 and who spent their childhood in a county, we must consider carefully the years of MSA-estimates of economic freedom to use. Indeed, using an economic freedom index for 2017 is not going to be relevant as its economic freedom up to adulthood that we seek to study. We also care about the economic freedom of parents just before they have children. As such, we average values of economic freedom in each MSA from 1977 up to 2017. This way we produce a “lifetime” estimate of the exposure to economically free environment that a person enjoys. In our supplementary materials, we provide an alternative set of results with an economic freedom index “childhood average” that covers the years between the closest to 1978 and the closest to 2001 (such that the birth cohorts from 1978 to 1983 is 18 years older between 1996 and 2001). Our results are conceptually similar and show the same pattern of significance across these differing definitions of “lifetime” economic freedom.

As our data is cross-sectional in nature – just as for Chetty et al. (2022a) – we are constrained to use an Ordinary Least Squares (OLS) strategy of the following type.

$$1) M_i = MEFI_i\beta_1 + SK_i\beta_2 + X'\delta + e$$

where  $M_i$  is one of the measures of income mobility (i.e., absolute mobility, relative mobility, racial in absolute mobility),  $MEFI_i$  is one of the measures of economic freedom (i.e., the aggregate index and its three components),  $SK_i$  is one of the measures of social capital (i.e., economic connectedness, clustering, support ratio, civic organizations and volunteering rates).<sup>18</sup> A matrix of control variables  $X'$  is included. Informing our choice of control variables is the work of Meehan et al.

(2019) who used the data of Chetty et al. (2014) to test the effect of the growth of licensing of low-income occupations at the county-level on income mobility: teenage pregnancies,<sup>19</sup> human capital<sup>20</sup>, proportion of uninsured individuals below 65 years old, county-level inequality (which we label  $I_i$  for reasons that we discuss below)<sup>21</sup> and the average age of population in the county. All of these are available from the American Community Survey (ACS) for 2019 (avoiding issues of non-response biases during COVID-era surveys). In supplementary materials, we provide Table S1 explaining all variables in greater details. We include a state dummy to account for differences in state-level policies or unmeasured differences in state-level characteristics.

To test whether economic freedom is a substitute or complement to the different types of social capital and whether inequality’s effects are mitigated/amplified by economic freedom we produce variants on this specification 1) by adding interaction terms in the form.

$$2) M_i = MEFI_i\tilde{\beta}_1 + SK_i\tilde{\beta}_2 + (MEFI_i \cdot SK_i)\tilde{\beta}_3 + X'\tilde{\delta} + e$$

$$3) M_i = MEFI_i\tilde{\beta}_1 + I_i\tilde{\beta}_2 + (MEFI_i \cdot I_i)\tilde{\beta}_3 + X'\tilde{\delta} + e$$

where the variables are the same as in 1) but with the exception that we include interaction terms with either income inequality or any of the measures of social capital. For conciseness, we only present the results from specifications 2) and 3) with the aggregate index of economic freedom. Presenting the results with each of the three subcomponents would multiply the number of tables by three which would make this article unwieldy. We refer readers to our supplementary materials.

We conduct a series of robustness tests. In a supplementary file, we swap between control variables and add new ones, switch to sub-components of the three economic freedom components, switch to LASSO regressions, limit the number of years of economic freedom data, switch to alternative measures of pro-market policies (see more in discussion), shift to other measures of social capital that Chetty et al. (2022) did not use. This is more than 440 additional specifications on

<sup>19</sup> Representing family stability for a person.

<sup>20</sup> Percentage with more than high school degree.

<sup>21</sup> Chetty et al.’s Gini coefficient is estimated using tax data minus the income share of the top 1 %. As such, it is a measure of inequality among the bottom 99 % in each county.

<sup>18</sup> In our supplementary materials, we will also replicate our results with all the measures of social capital included together.

top of those in the present article. None of these checks change our results which is why we place them in a lengthy supplementary material scml file.

Two of the most important checks we use relate to potential endogeneity and the familywise error rate stemming from the use of economic freedom as an aggregate and then shifting to the components of the index. To deal with the former, we use instrumental approaches. Potential endogeneity between economic mobility and institutions emerge over concerns that low economic mobility could trigger institutional reforms that lead to changes in mobility. We are skeptical of such a link as it implies strong foresight from parents regarding the income mobility they expect for their children (and that they care sufficiently to be correctly informed about policy effects). Nevertheless, to mitigate concerns over simultaneity to the extent possible, we will re-estimate the main results using two instrumental variables strategies. The first uses each the pre-sample (1972) value of each economic freedom component as an external instrument. The second combines each external instrument with a data-driven instrument following the approach proposed by [Lewbel \(2012\)](#), which relies on higher-order moment restrictions from the first-stage for identification (i.e., heteroskedasticity) rather than the traditional exclusion restriction associated with external instruments ([Lewbel, 2012](#)). Combining a Lewbel instrument with the pre-sample economic freedom value provides us with a robustness check and allows us to also examine instrument exogeneity.<sup>22</sup>

With respect to the latter, shifting from the overall MEFI index to its three individual components necessarily involves multiple hypothesis testing. By evaluating several related hypotheses (aggregate economic freedom and the separate effects of the three components), there is the possibility of falsely rejecting the null hypothesis simply by virtue of the number of tests conducted ([Benjamini and Hochberg, 1995](#)). To address this issue, we employ the Holm-Bonferroni procedure ([Abdi, 2010](#)). It mitigates the concern in a stepwise manner whereby we rank the  $p$ -values from all related tests from smallest to largest. The smallest  $p$ -value is compared to the strictest threshold ( $\frac{\alpha}{m}$  where  $\alpha$  is the chosen significance level and  $m = 4$  is the number of tests). If it passes the tests, the next smallest is compared to against a slightly less strict threshold:  $\frac{\alpha}{m-1}$ . This goes on until one test fails and all subsequent estimates are deemed not significant at the  $\alpha$  level. For consistency and transparency, we report all such results in each regression where economic freedom and its components are used (this means [Tables 2, 4–8](#) and 11). All tests where only the aggregate MEFI is used do not require the use of Holm-Bonferroni procedure. We use  $\alpha = 0.05$ .

### 3. Results

To measure different facets of economic freedom, we use an aggregated index of economic freedom, proxies for the government spending, taxation, and labor market freedom. These measures are developed using several subindices (as we mention above in the data section) to measure aspects like government consumption, transfers and subsidies, government investment, top marginal tax rates, as well as income and payroll taxes. While we use the aggregate measures for brevity, all models are repeated for each subindex in the reference file.

<sup>22</sup> Lewbel's paper has also been cited more 2000 times, evidence of its widespread acceptance as a valid method of identification. [Lewbel's \(2012\)](#) data-driven, or synthetic instrument, approach is based on the assumption of multiplicative heteroskedasticity. Using a Breusch-Pagan test, we rejected the null hypothesis of homoskedasticity in each specification, lending support for Lewbel's method. Additionally, we also follow [Lee et al. \(2022\)](#) and have adjusted the t-statistics and p-values of the endogenous variables based on the strength of the instruments (first-stage F), which vary between 13.5 and 44.7. P-values increased by roughly 20 % after adjusting for instrument strength. We have omitted reporting the first-stage F statistics because the correction procedure integrates this information into the reported t-statistics and p-values.

**Table 1**

Summary statistics.

| Variable                         | Obs | Mean   | Std. dev. | Min    | Max    |
|----------------------------------|-----|--------|-----------|--------|--------|
| <i>Economic Freedom Measures</i> |     |        |           |        |        |
| Economic Freedom                 | 261 | 6.473  | 0.784     | 4.071  | 8.113  |
| Size of Govt (Area 1)            | 261 | 6.922  | 1.081     | 2.555  | 8.694  |
| Taxation (Area 2)                | 261 | 6.013  | 0.744     | 3.401  | 7.888  |
| Labor Mkt Free. (Area 3)         | 261 | 6.484  | 0.919     | 4.079  | 8.350  |
| <i>Social Capital Measures</i>   |     |        |           |        |        |
| Economic Conn.                   | 261 | 0.809  | 0.140     | 0.491  | 1.220  |
| Clustering                       | 261 | 0.093  | 0.009     | 0.076  | 0.120  |
| Support Ratio                    | 261 | 0.981  | 0.013     | 0.915  | 0.998  |
| Volunteering                     | 261 | 0.070  | 0.023     | 0.028  | 0.149  |
| Civic Org.                       | 261 | 0.015  | 0.004     | 0.004  | 0.031  |
| <i>Mobility Measures</i>         |     |        |           |        |        |
| Absolute Mobility                | 261 | 41.543 | 3.439     | 33.728 | 51.190 |
| Relative Mobility                | 261 | 0.674  | 0.056     | 0.566  | 0.830  |
| White-Black Abs. Mob             | 261 | 0.112  | 0.046     | −0.048 | 0.469  |

Note: We invert the relative mobility score from [Chetty et al. \(2014\)](#) so that higher scores of relative mobility corresponds to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile.

**Table 2**

Summary of Economic Freedom on Mobility (Including Controls, without social capital).

|                                                  | Absolute Mob.   | Relative Mob.   | White-Black Gap in Absolute Mob. |
|--------------------------------------------------|-----------------|-----------------|----------------------------------|
| Economic Freedom                                 | 2.154***        | 0.018**         | 0.002                            |
| Size of Govt (Area 1)                            | 1.421***        | 0.013***        | −0.003                           |
| Taxation (Area 2)                                | 0.779           | 0.009           | −0.011                           |
| Labor Market Freedom (Area 3)                    | 1.626***        | 0.010*          | 0.013*                           |
| Holm-Bonferroni Adjustment (at $\alpha = 0.05$ ) |                 |                 |                                  |
| Economic Freedom                                 | Significant     | Significant     | Not significant                  |
| Size of Govt (Area 1)                            | Significant     | Significant     | Not significant                  |
| Taxation (Area 2)                                | Not significant | Not significant | Not significant                  |
| Labor Market Freedom (Area 3)                    | Significant     | Not significant | Not significant                  |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from [Chetty et al. \(2014\)](#), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile. The Holm-Bonferroni method that controls for family-wise error rate by ranking the p-values from smallest to largest and then testing them sequentially against progressively less strict thresholds ( $\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha$ ) where  $\alpha$  is the significance level and  $m$  is the number of separate tests on family-wise components (here  $m = 4$  because the set is Economic Freedom, Area 1, Area 2, Area 3).

[Table 1](#) provides summary statistics of both economic freedom and the social capital measures provided by [Chetty et al. \(2022a\)](#). There are some notable level differences, as economic freedom and its subindices are measured on a scale between 1 and 10, with most scores lying between 3 and 8. Social capital measures, discussed above and provided through Chetty's publicly available files, are scored between 0 and 1 or most subindices. The differences in scaling will be relevant when discussing the magnitude of relationships with intergenerational mobility and relative effect sizes.

**Table 3**

Social Capital on Mobility (Including Controls, with social capital and without economic freedom).

|                        | Absolute Mob. | Relative Mob. | White-Black Gap in Absolute Mob. |
|------------------------|---------------|---------------|----------------------------------|
| Economic Connectedness | 16.755***     | 0.219***      | −0.028                           |
| Clustering             | −6.542        | −0.658*       | 1.102**                          |
| Support Ratio          | −34.357**     | −0.757***     | −0.146                           |
| Volunteering Rate      | 4.683         | −0.009        | −0.214                           |
| Civic Organizations    | 0.013         | −0.049        | −0.129                           |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile.

Descriptive statistics for our dependent variables of interest (absolute, relative, and gap mobility) are presented in the third panel of Table 1. It should be noted that relative mobility, taken from Chetty et al. (2014), is inverted for ease of interpretability so that positive coefficient estimates correspond to higher levels of relative mobility. To ensure that economic freedom and social capital are not colinear or accounting for similar variation, we present a series of scatter plots of the measure types in Fig. 3, showing that economic freedom and social capital are largely uncorrelated.<sup>23</sup>

Prior to combining the measures, we also want to understand the relationship between the measures and our dependent variables separately. This allows us to assess two crucial areas of concern. The first is understanding the baseline relationship between economic freedom and mobility, as well as between social capital and mobility, when not accounting for economic freedom. The second is to provide an identical specification to the county-level assessment between the social capital measures and both aggregate and relative mobility provided by Chetty et al. (2022), to ensure our MSA-level analysis is consistent with theirs.

Table 2 regresses the four economic freedom measures against absolute, relative, and gap mobility. Note that each cell within the table represents a separate regression of one measure of economic freedom against one measure of economic mobility, resulting in twelve separate regressions. For the sake of brevity, we do not report the coefficient estimates of the control variables (state-level fixed effects, average age, percent with more than a high school degree, teen birth rate, AGI Gini coefficient, and rate of uninsured persons), but each of the full models is available in the supplementary materials. We find that the composite measure of overall economic freedom is related with greater absolute and relative mobility, while having no noticeable effect on the White-Black gap in relative mobility (suggesting that the gains are equally experienced across racial groups). To put things into perspective, a one standard deviation higher level of economic freedom predicts absolute mobility being higher by half a standard deviation. For relative mobility, that proportion is one quarter of a standard deviation. When decomposing into the three main areas of economic freedom, we find that government spending and labor market freedoms are associated with greater absolute and relative mobility. However, relaxations in labor market regulations appear to be weakly associated with better outcomes for Whites. Government taxation policies do not appear to influence

**Table 4**

Effect of economic freedom and economic connectedness on mobility.

| Panel A: Absolute Mobility                       |                  |              |             |                    |
|--------------------------------------------------|------------------|--------------|-------------|--------------------|
|                                                  | (1)              | (2)          | (3)         | (4)                |
| Economic Freedom                                 | 1.810***         |              |             |                    |
| Size of Govt (Area 1)                            |                  | 1.098***     |             |                    |
| Taxation (Area 2)                                |                  |              | 0.867**     |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |             | 1.392***           |
| Economic Connectedness                           | 16.077***        | 15.885***    | 16.823***   | 16.185***          |
| Panel B: Relative Mobility                       |                  |              |             |                    |
|                                                  | (5)              | (6)          | (7)         | (8)                |
| Economic Freedom                                 | 0.014**          |              |             |                    |
| Size of Govt (Area 1)                            |                  | 0.009**      |             |                    |
| Taxation (Area 2)                                |                  |              | 0.010       |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |             | 0.007              |
| Economic Connectedness                           | 0.214***         | 0.212***     | 0.220***    | 0.216***           |
| Panel C: White-Black Gap                         |                  |              |             |                    |
|                                                  | (9)              | (10)         | (11)        | (12)               |
| Economic Freedom                                 | 0.002            |              |             |                    |
| Size of Govt (Area 1)                            |                  | −0.003       |             |                    |
| Taxation (Area 2)                                |                  |              | −0.011      |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |             | 0.013*             |
| Economic Connectedness                           | −0.029           | −0.026       | −0.029      | −0.034             |
| Holm-Bonferroni Adjustment (at $\alpha = 0.05$ ) |                  |              |             |                    |
|                                                  | Economic Freedom | Size of Govt | Taxation    | Labor Mrkt Freedom |
| Absolute Mobility                                | Significant      | Significant  | Significant | Significant        |
| Relative Mobility                                | Not              | Not          | Not         | Not                |
| White-Black Gap                                  | Significant      | Significant  | Significant | Significant        |
|                                                  | Not              | Not          | Not         | Not                |
|                                                  | Significant      | Significant  | Significant | Significant        |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile. The Holm-Bonferroni method that controls for family-wise error rate by ranking the p-values from smallest to largest and then testing them sequentially against progressively less strict thresholds  $(\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha)$  where  $\alpha$  is the significance level and  $m$  is the number of separate tests on family-wise components (here  $m = 4$  because the set is Economic Freedom, Area 1, Area 2, Area 3).

<sup>23</sup> In supplementary materials, we also conducted tests using variance inflation factors between economic freedom and social capital measures. None of the factors returned are above five indicating minimal issues of collinearity.



**Table 5**  
Effect of economic freedom and clustering on mobility.

| Panel A: Absolute Mobility  |                                                  |                 |                 |                    |
|-----------------------------|--------------------------------------------------|-----------------|-----------------|--------------------|
|                             | (1)                                              | (2)             | (3)             | (4)                |
| Economic Freedom            | 2.152***                                         |                 |                 |                    |
| Size of Govt (Area 1)       |                                                  | 1.429***        |                 |                    |
| Taxation (Area 2)           |                                                  |                 | 0.779           |                    |
| Labor Mrkt Freedom (Area 3) |                                                  |                 |                 | 1.639***           |
| Clustering                  | −3.213                                           | −12.120         | −6.366          | 7.087              |
| Panel B: Relative Mobility  |                                                  |                 |                 |                    |
|                             | (5)                                              | (6)             | (7)             | (8)                |
| Economic Freedom            | 0.018**                                          |                 |                 |                    |
| Size of Govt (Area 1)       |                                                  | 0.014***        |                 |                    |
| Taxation (Area 2)           |                                                  |                 | 0.009           |                    |
| Labor Mrkt Freedom (Area 3) |                                                  |                 |                 | 0.009*             |
| Clustering                  | −0.631*                                          | −0.713*         | −0.656*         | −0.580             |
| Panel C: White-Black Gap    |                                                  |                 |                 |                    |
|                             | (9)                                              | (10)            | (11)            | (12)               |
| Economic Freedom            | 0.002                                            |                 |                 |                    |
| Size of Govt (Area 1)       |                                                  | −0.004          |                 |                    |
| Taxation (Area 2)           |                                                  |                 | −0.010          |                    |
| Labor Mrkt Freedom (Area 3) |                                                  |                 |                 | 0.015**            |
| Clustering                  | 1.106**                                          | 1.117**         | 1.100**         | 1.228**            |
|                             | Holm-Bonferroni Adjustment (at $\alpha = 0.05$ ) |                 |                 |                    |
|                             | Economic Freedom                                 | Size of Govt    | Taxation        | Labor Mrkt Freedom |
| Absolute Mobility           | Significant                                      | Significant     | Not Significant | Significant        |
| Relative Mobility           | Significant                                      | Significant     | Not Significant | Not Significant    |
| White-Black Gap             | Not Significant                                  | Not Significant | Not Significant | Significant        |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile. The Holm–Bonferroni method that controls for familywise error rate by ranking the p-values from smallest to largest and then testing them sequentially against progressively less strict thresholds  $\left(\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha\right)$  where  $\alpha$  is the significance level and  $m$  is the number of separate tests on family-wise components (here  $m = 4$  because the set is Economic Freedom, Area 1, Area 2, Area 3).

economic mobility. While not reported in Table 2 for conciseness (but available in the supplementary materials), we also find that the Gini coefficient is generally weaker than the effect of economic freedom.<sup>24</sup> The pattern of statistical significance is similar with the use of the Holm–Bonferroni procedure with the exception of labor market freedom (with respect to relative income mobility and the gap in mobility).

An identical analysis is repeated for all the publicly available measures of social capital used by Chetty et al. (2022) in Table 3. The direction and significance of these measures on economic mobility closely match the coefficient estimates reported in their work. As in Table 2, each cell represents a separate equation. Greater economic connectedness is related with greater mobility, while a higher support ratio is associated with lower mobility. Coefficient estimates for clustering and civic organizations replicate the results of Chetty et al. (2022) in terms of both sign and insignificance. Volunteering shows a similar magnitude and direction, but we did not find the statistically significant relationship reported by Chetty et al. (2022). These results reinforce our confidence that our population-weighted method for aggregating to the MSA level preserves the same relationships and patterns observed in the

original paper.

In Table 4 through 8, we isolate each pair of economic freedom and social capital variables and regress both against our three measures of economic mobility to assess their relative importance. Table 4, which is the most important given the emphasis of Chetty et al. (2022a) on economic connectedness, estimates the relationship between economic connectedness and the four components of economic freedom across our three mobility measures. Departing from the format used in Tables 2 and 3, each number-panel column in Table 4 represents a separate regression. For absolute mobility, higher scores in all measures of economic freedom are associated with higher mobility when economic connectedness is included. For relative mobility, greater economic freedom and lower government spending are significantly associated with greater mobility. Taxation and labor market regulation have the right same sign but they are not significantly different from zero. Regarding the White-Black mobility gap, we do not find evidence that the presence of both economic freedom and economic connectedness systematically benefits either group. There is some indication that increases in labor market freedom are associated with a modest widening of the White-Black mobility gap. Expressed in standard deviation units, we find that a one standard deviation increase in economic connectedness raises absolute mobility by 65 % of a standard deviation and relative mobility by 54 %. A one standard deviation increase in economic freedom is associated with higher absolute mobility by 41 % of a standard deviation, and in relative mobility by 20 %. Across all tables, the pattern of statistical significance is similar with the use of the Holm–Bonferroni procedure.

This analysis is repeated for each of the other social capital measures

<sup>24</sup> We also note that, in the supplementary materials, we attempt to account for the multiplicative role of the components of economic freedom, as emphasized by Bolen and Sobel (2020). To do so, we standardize the components and find that the effect of economic freedom is still strong. In other words, when holding the degree of unevenness across components constant, higher levels of economic freedom are associated with slightly greater improvements in income mobility. The same is true for Table 4 below.

**Table 6**  
Effect of economic freedom and support ratio on mobility.

| Panel A: Absolute Mobility                       |                  |              |                 |                    |
|--------------------------------------------------|------------------|--------------|-----------------|--------------------|
|                                                  | (1)              | (2)          | (3)             | (4)                |
| Economic Freedom                                 | 2.043***         |              |                 |                    |
| Size of Govt (Area 1)                            |                  | 1.388***     |                 |                    |
| Taxation (Area 2)                                |                  |              | 0.796*          |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |                 | 1.503***           |
| Support Ratio                                    | −25.634*         | −31.108**    | −34.681**       | −18.007            |
| Panel B: Relative Mobility                       |                  |              |                 |                    |
|                                                  | (5)              | (6)          | (7)             | (8)                |
| Economic Freedom                                 | 0.015**          |              |                 |                    |
| Size of Govt (Area 1)                            |                  | 0.013***     |                 |                    |
| Taxation (Area 2)                                |                  |              | 0.009           |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |                 | 0.006              |
| Support Ratio                                    | −0.692***        | −0.727***    | −0.761***       | −0.695***          |
| Panel C: White-Black Gap                         |                  |              |                 |                    |
|                                                  | (9)              | (10)         | (11)            | (12)               |
| Economic Freedom                                 | 0.001            |              |                 |                    |
| Size of Govt (Area 1)                            |                  | −0.003       |                 |                    |
| Taxation (Area 2)                                |                  |              | −0.010          |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |                 | 0.013*             |
| Support Ratio                                    | −0.141           | −0.154       | −0.142          | −0.006             |
| Holm-Bonferroni Adjustment (at $\alpha = 0.05$ ) |                  |              |                 |                    |
|                                                  | Economic Freedom | Size of Govt | Taxation        | Labor Mrkt Freedom |
| Absolute Mobility                                | Significant      | Significant  | Not Significant | Significant        |
| Relative Mobility                                | Not Significant  | Significant  | Not Significant | Not Significant    |
| White-Black Gap                                  | Significant      | Significant  | Significant     | Significant        |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile. The Holm-Bonferroni method that controls for family-wise error rate by ranking the p-values from smallest to largest and then testing them sequentially against progressively less strict thresholds  $(\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha)$  where  $\alpha$  is the significance level and  $m$  is the number of separate tests on family-wise components (here  $m = 4$  because the set is Economic Freedom, Area 1, Area 2, Area 3).

from Chetty et al. (2022), including clustering (Table 5), support ratio (Table 6), volunteering rate (Table 7), and the presence of civic organizations (Table 8). Unlike economic connectedness, economic freedom consistently appears stronger—and more beneficial—than all these alternative measures of social capital. When controlling for clustering,

**Table 7**  
Effect of economic freedom and volunteering rate on mobility.

| Panel A: Absolute Mobility                       |                  |              |                 |                    |
|--------------------------------------------------|------------------|--------------|-----------------|--------------------|
|                                                  | (1)              | (2)          | (3)             | (4)                |
| Economic Freedom                                 | 2.199***         |              |                 |                    |
| Size of Govt (Area 1)                            |                  | 1.425***     |                 |                    |
| Taxation (Area 2)                                |                  |              | 0.810*          |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |                 | 1.689***           |
| Volunteering Rate                                | 8.3015           | 5.339        | 5.797           | 9.949              |
| Panel B: Relative Mobility                       |                  |              |                 |                    |
|                                                  | (5)              | (6)          | (7)             | (8)                |
| Economic Freedom                                 | 0.018**          |              |                 |                    |
| Size of Govt (Area 1)                            |                  | 0.014***     |                 |                    |
| Taxation (Area 2)                                |                  |              | 0.009           |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |                 | 0.011*             |
| Volunteering Rate                                | 0.021            | −0.003       | −0.003          | 0.024              |
| Panel C: White-Black Gap                         |                  |              |                 |                    |
|                                                  | (9)              | (10)         | (11)            | (12)               |
| Economic Freedom                                 | 0.0004           |              |                 |                    |
| Size of Govt (Area 1)                            |                  | −0.003       |                 |                    |
| Taxation (Area 2)                                |                  |              | −0.012          |                    |
| Labor Mrkt Freedom (Area 3)                      |                  |              |                 | 0.012*             |
| Volunteering Rate                                | −0.213           | −0.216       | −0.230          | −0.178             |
| Holm-Bonferroni Adjustment (at $\alpha = 0.05$ ) |                  |              |                 |                    |
|                                                  | Economic Freedom | Size of Govt | Taxation        | Labor Mrkt Freedom |
| Absolute Mobility                                | Significant      | Significant  | Not Significant | Significant        |
| Relative Mobility                                | Significant      | Significant  | Not Significant | Not Significant    |
| White-Black Gap                                  | Significant      | Significant  | Significant     | Significant        |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile. The Holm-Bonferroni method that controls for family-wise error rate by ranking the p-values from smallest to largest and then testing them sequentially against progressively less strict thresholds  $(\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha)$  where  $\alpha$  is the significance level and  $m$  is the number of separate tests on family-wise components (here  $m = 4$  because the set is Economic Freedom, Area 1, Area 2, Area 3).

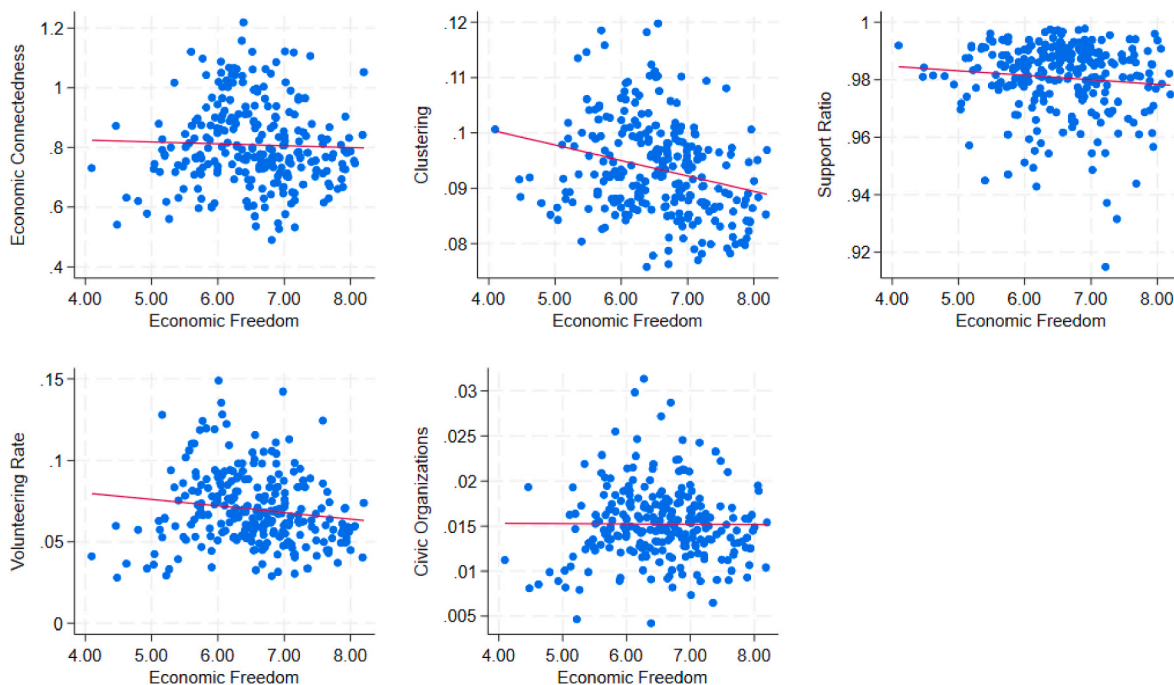
higher overall economic freedom, lower government spending, and greater labor market freedom continue to show positive effects on both absolute and relative mobility, whereas clustering depresses mobility (albeit with an inconsistent pattern of statistical significance). Interestingly, while higher economic freedom is not associated with substantial differences in mobility between White and Black individuals, higher clustering is strongly associated with greater improvements in mobility

**Table 8**

Effect of economic freedom and civic organizations on mobility.

| Panel A: Absolute Mobility  |                                                  |                 |                 |                    |
|-----------------------------|--------------------------------------------------|-----------------|-----------------|--------------------|
|                             | (1)                                              | (2)             | (3)             | (4)                |
| Economic Freedom            | 2.161***                                         |                 |                 |                    |
| Size of Govt (Area 1)       |                                                  | 1.422***        |                 |                    |
| Taxation (Area 2)           |                                                  |                 | 0.780           |                    |
| Labor Mrkt Freedom (Area 3) |                                                  |                 |                 | 1.638***           |
| Civic Organizations         | 11.647                                           | 3.741           | 2.554           | 16.841             |
| Panel B: Relative Mobility  |                                                  |                 |                 |                    |
|                             | (5)                                              | (6)             | (7)             | (8)                |
| Economic Freedom            | 0.018**                                          |                 |                 |                    |
| Size of Govt (Area 1)       |                                                  | 0.014***        |                 |                    |
| Taxation (Area 2)           |                                                  |                 | 0.009           |                    |
| Labor Mrkt Freedom (Area 3) |                                                  |                 |                 | 0.011*             |
| Civic Organizations         | 0.050                                            | −0.013          | −0.020          | 0.059              |
| Panel C: White-Black Gap    |                                                  |                 |                 |                    |
|                             | (9)                                              | (10)            | (11)            | (12)               |
| Economic Freedom            | 0.002                                            |                 |                 |                    |
| Size of Govt (Area 1)       |                                                  | −0.003          |                 |                    |
| Taxation (Area 2)           |                                                  |                 | −0.011          |                    |
| Labor Mrkt Freedom (Area 3) |                                                  |                 |                 | 0.013*             |
| Civic Organizations         | −0.121                                           | −0.137          | −0.164          | 0.003              |
|                             | Holm-Bonferroni Adjustment (at $\alpha = 0.05$ ) |                 |                 |                    |
|                             | Economic Freedom                                 | Size of Govt    | Taxation        | Labor Mrkt Freedom |
| Absolute Mobility           | Significant                                      | Significant     | Not Significant | Significant        |
| Relative Mobility           | Significant                                      | Significant     | Not Significant | Not Significant    |
| White-Black Gap             | Not Significant                                  | Not Significant | Not Significant | Not Significant    |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile. The Holm–Bonferroni method that controls for familywise error rate by ranking the p-values from smallest to largest and then testing them sequentially against progressively less strict thresholds  $\left(\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha\right)$  where  $\alpha$  is the significance level and  $m$  is the number of separate tests on family-wise components (here  $m = 4$  because the set is Economic Freedom, Area 1, Area 2, Area 3).

**Fig. 3.** Scatter plots of economic freedom on measures of social capital.

**Table 9**  
Effect of economic freedom and social capital on mobility (interaction term).

| Panel A: Absolute Mobility        |            |            |         |              |           |
|-----------------------------------|------------|------------|---------|--------------|-----------|
|                                   | Econ Conn. | Clustering | Support | Volunteering | Civic Org |
| EF*SC                             | −0.046     | −47.951**  | 4.266   | −3.011       | −94.513** |
| Panel B: Relative Mobility        |            |            |         |              |           |
|                                   | Econ Conn. | Clustering | Support | Volunteering | Civic Org |
| EF*SC                             | 0.026      | −1.325***  | −0.253  | 0.232*       | −0.577    |
| Panel C: White-Black Mobility Gap |            |            |         |              |           |
|                                   | Econ Conn. | Clustering | Support | Volunteering | Civic Org |
| EF*SC                             | 0.033      | 0.237      | 0.226   | 0.170        | 0.733     |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile.

**Table 10**  
Effect of economic freedom and social capital on mobility (interaction term).

| Panel A: Absolute Mobility        |            |            |         |              |           |
|-----------------------------------|------------|------------|---------|--------------|-----------|
|                                   | Econ Conn. | Clustering | Support | Volunteering | Civic Org |
| Inequality * EF                   | 23.697     | 41.674     | 48.140  | 43.727       | 42.140    |
| Panel B: Relative Mobility        |            |            |         |              |           |
|                                   | Econ Conn. | Clustering | Support | Volunteering | Civic Org |
| Inequality * EF                   | −0.259     | −0.060     | 0.146   | −0.011       | −0.015    |
| Panel C: White-Black Mobility Gap |            |            |         |              |           |
|                                   | Econ Conn. | Clustering | Support | Volunteering | Civic Org |
| Inequality * EF                   | 1.086      | 1.130      | 1.091   | 1.005        | 1.047     |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included (all at the MSA-level). Economic freedom scores and its areas are the average from 1972 to 2017. Note that each coefficient presented is from a separate regression (therefore, 12 in total). We invert the relative mobility score from Chetty et al. (2014), so that higher scores of relative mobility correspond to higher levels of mobility. White-Black gap in absolute mobility is the average rank of a white person born in the 25th percentile minus the average rank of a black person born in the 25th percentile.

for White families relative to Black families. This suggests that clustering may amplify racial inequality, potentially reflecting the role of privileged or “elite” groups in capturing excess rents through exclusive networks. While increases in economic freedom remain consistent across several models and tables, increases in the support ratio are weakly associated with lower absolute mobility and strongly associated with reduced relative mobility, though they do not contribute to widening the racial or ethnic mobility gap. Volunteering and the presence of civic organizations show no meaningful relationship with any form of economic mobility once economic freedom is accounted for.

While we have evidence of minimal collinearity between our measures of economic freedom and social capital, and likewise between economic freedom and inequality, determining whether their effects are

**Table 11**  
Instrumental variables estimates from Table 2.

|                                              | Absolute Mob. | Relative Mob. | White-Black Gap in Absolute Mob. |
|----------------------------------------------|---------------|---------------|----------------------------------|
| Economic Freedom                             |               |               |                                  |
| 1972 value as instrument                     | 1.716***      | 0.034***      | −0.001                           |
| 1972 value as instrument + Lewbel instrument | 1.674***      | 0.033***      | −0.001                           |
| Sargan p-value                               | 0.275         | 0.074         | 0.705                            |
| Size of Govt (Area 1)                        |               |               |                                  |
| 1972 value as instrument                     | 1.348***      | 0.027***      | −0.0004                          |
| 1972 value as instrument + Lewbel instrument | 1.437***      | 0.027***      | 0.0002                           |
| Sargan p-value                               | 0.179         | 0.505         | 0.676                            |
| Taxation (Area 2)                            |               |               |                                  |
| 1972 value as instrument                     | 2.541         | 0.050**       | −0.001                           |
| 1972 value as instrument + Lewbel instrument | 2.019         | 0.042**       | −0.008                           |
| Sargan p-value                               | 0.078         | 0.082         | 0.194                            |
| Labor Market Freedom (Area 3)                |               |               |                                  |
| 1972 value as instrument                     | 1.634***      | 0.032***      | −0.0005                          |
| 1972 value as instrument + Lewbel instrument | 1.639***      | 0.029**       | 0.0003                           |
| Sargan p-value                               | 0.956         | 0.049         | 0.640                            |

Notes: \*\*\*, \*\*, \* indicate significance at the 0.01, 0.05, and 0.10 level, respectively. P-values (and significance levels) are based on the valid t-ratio for instrumental variables regressions proposed by Lee et al. (2022). For each regression, the Lewbel instrument was formed using the uninsured rate. The Sargan p-value is from the overidentifying restrictions test.

Full results from the regressions can be found in the replication file. State-level fixed effects, average age, percent with more than a high school degree, teen birth rate, Gini coefficients, uninsured rate are included. With Holm-Bonferroni p-values, the hypothesis tests yield the same pattern of statistical significance with  $\alpha = 0.05$ . Ergo, reporting them would look redundant.

additive or substitutive requires assessing whether we are overlooking any interaction effects. Specifically, we must account for the possibility that areas with high economic freedom may also systematically exhibit higher levels of social capital or greater income inequality. To explore this, Table 9 presents interaction terms between economic freedom and each measure of social capital, using only the aggregate economic freedom index for brevity. Table 10 replicates this approach for income inequality. The component analysis for government spending, taxation and labor market freedom generally follows similar patterns, while taxation does not show any consistent effects. These models are available in the replication files and supplementary materials.

Overall, when accounting for potential interactions, increases in economic freedom continue to improve absolute mobility across all measures of social capital. The interaction between economic freedom and economic connectedness is not statistically significant, suggesting the two variables operate independently and are likely additive. However, interactions with clustering negatively predict mobility, implying that certain forms of social capital may act as substitutes for economic freedom. This supports the idea that social capital, in some forms, reduce the marginal benefit of economic freedom. Importantly, economic freedom does not appear to predict racial disparities in mobility—as measured by the Black–White mobility gap—under any specification. Taken together, the evidence suggests that the impact of economic freedom and “bridging” social capital are additive rather than complementary or substitutive. In contrast, the results for “bonding social capital” – proxied by clustering – align more closely with the view that economic freedom can serve as a substitute for social capital because it potentially promotes rent-seeking (Mathers and Williamson, 2011; Williamson and Mathers, 2011; Holcombe, 2018). In Table 10, the interaction between economic freedom and income inequality yields no significant effects, regardless of which social capital measure is included. This suggests that economic freedom does not exert its influence on mobility by moderating or amplifying the effects of



inequality—a possibility raised in some of the literature linking economic freedom to distributional outcomes.

Finally, in Table 11, we deal with the potential concern over endogeneity. This is done by replicating our results from Table 2 but using the aforementioned instrumental strategies. When the coefficients are compared with those of Table 2, we can see that there are trivial differences between the estimators (suggesting that endogeneity was not an issue) and that pattern of significance is strong regardless of whether we employ the external instrument of the MEFI value for 1972 or the Lewbel instrument combined with the MEFI value for 1972. The pattern of statistical significance is identical with the use of the Holm-Bonferroni procedure (see footnote to Table 11).

#### 4. Discussion and conclusion

Our article is the first to consider the relationship between economic freedom and social mobility using intergenerational data for a large sample of observations in conjunction with high quality social capital data. It is also the first that focuses on intergenerational data in the context of a secure property rights environment – which means that the attention goes largely to the role of regulations, government spending and taxation. Finally, it is also the first to do by considering the great degree of regional heterogeneity within subnational (i.e., states, provinces) districts.

Our results suggests that the current of the literature – which we argue largely ignores the role of markets and broader institutions – needs to devote more attention to the role of economic freedom as its importance is far from negligible. It should invite policy makers to consider pro-market policies that increase competition, eliminate entry barriers, foster investment and promote entrepreneurship.

In fact, we believe that our results downplay the relevance of such policy actions. The reasons for this downplay can potentially inform more concrete policy proposals. First, the economic freedom index at the subnational level omits key regulations such as childcare regulations (which matter heavily for the labor supply decisions of mothers),<sup>25</sup> housing regulations, industry-level regulations, small business regulations, and occupational licensing.<sup>26</sup> The latter is particularly important, as occupational licensing has dramatically expanded in recent decades to cover an increasing share of low-to middle-income occupations (Carpenter et al., 2015, 2018). Licensing has also been linked to lower intergenerational income mobility (Meehan et al., 2019; Geloso et al., 2024) and poorer socio-economic outcomes for lower-income groups (Deyo and Plemmons, 2022; Plemmons, 2022; Plemmons and Timmons, 2023). Occasional measures of licensing are strongly correlated with economic freedom indexes in matching years. The problem, akin to that of property rights, is that the data are often only available at the state level or are not consistently available across multiple years. Yet, in our supplementary materials, we also attempted to use occupational licensing data from Carpenter et al. (2021),<sup>27</sup> which are available for a

<sup>25</sup> Childcare regulations, which are not typically included in economic freedom indexes but are well-correlated with them (Flowers et al., 2024), exemplify this issue. Regulations affecting the supply of childcare services disproportionately impact women, as they tend to raise prices (Gorry and Thomas, 2017; Flowers et al., 2024), thereby influencing the decision not to re-enter the workforce after pregnancy (Powell, 2002; Baker et al., 2008; Lefebvre and Merrigan, 2008).

<sup>26</sup> However, in our supplementary materials, we show that the economic freedom indexes we use correlate well with several other regulatory measures when available—such as land-use regulation, occupational licensing, industry-level regulation by state, and small business regulations (Winegarden, 2015). These measures are also associated with lower income mobility. Therefore, augmenting the economic freedom index with these components over the full period since 1972 would likely strengthen the results presented in this article.

<sup>27</sup> A dataset recently employed by Plemmons and Ghosh (2022) in assessing between-city migration.

single year at the MSA level, to augment the index. We assess whether any results we obtain using the index of economic freedom—particularly the component related to labor market regulation—are conceptually echoed by this alternative measure. We find that our main results hold and are slightly stronger. Overall, it seems that the structure of the data used to build subnational economic freedom indexes likely understates the true importance of economic freedom. This, we believe, should be a call to augment these indexes by incorporating more data on components that are especially relevant to the income mobility of individuals near the bottom of the income ladder. At the same time, this highlights the practical relevance of concrete proposals such as occupational deregulation.

Second, the index of economic freedom we use does not include a property rights component since it was constructed with the assumption that subnational variations in property rights were too small to matter. However, Murphy (2020) showed that this assumption was somewhat incorrect by augmenting – unfortunately for a single year only – the index for subnational variations in property rights. Incorporating his result imposes a heavy trade-off data-wise for us because he is using a single year at the end of our window of income mobility and the variable is at the state level not the MSA-level. Yet, in our supplementary materials, we replicate our results but swap the state dummy variables for the state-level property rights measure constructed by Murphy. Our results show that the property rights considerations are strongly and positively correlated with absolute and relative mobility.

The question now is how to move to more causal assessments. Recent empirical studies of the connection between economic freedom and economic growth offer viable templates. For example, Grier and Grier (2021) use matching methods to assess whether “big jumps” in economic freedom indices are tied with faster rates of growth. Using international data, they found that big liberalization meant that countries had incomes per capita 16 % higher than the counterfactuals suggest. Callais and Young (2023) used the same method for the income of people at different deciles and their share of total income – finding that there were gains for all (if unevenly). Similar methods could be used for countries – like Canada or the United States – that have subnational data on intergenerational income mobility to assess what happens when a subnational government enacts large reforms. As way of another example, synthetic control methods could be used to assess counterfactual scenarios of changes in economic freedom by governments who improve property rights, deregulate occupations or reduce trade barriers.

Regardless of the way forward in terms of causal approaches, we see our results as a call to research that should be heeded before we find ourselves discussing arcane points of minute policymaking. Promoting open markets appear to remain a key ingredient for mobility.

#### Declaration of competing interest

The authors report no conflicts of interest.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.econmod.2025.107351>.

## Data availability

Data will be made available on request.

Institutions Matter: Economic Freedom and Income Mobility  
(Original data) (Mendeley Data)

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